

MAS-ESS Essay Competition - Arin Jain

Singapore's AI-Driven Economic Future: How can artificial intelligence reshape our industries, workforce, and competitive advantage while ensuring inclusive growth and managing disruption?

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Resources

- Winning sample: [Building Singapore's Digital Asset Ecosystem](#)
- 2000 Words Maximum
- Terms and Conditions: [2026 Essay Comp Terms and Conditions.pdf](#)

Summary

Artificial intelligence is no longer the distant prospect we thought it was; it now has become a present economic force, reshaping industries, labour markets, and market behaviour worldwide. For Singapore—a small, ageing and relatively high-cost advanced economy with a trade-to-GDP ratio above 300%—how AI is utilized and adopted will determine whether the various benefits it poses are broad or concentrated among the few.

To address this, this essay surveys Singapore’s current AI activities across three domains—industry (finance, logistics, and healthcare), the broader workforce, and its positioning as a regional hub—evaluating both the productivity gains being delivered and the risks which accumulate (such as SME exclusion and strategic over-dependence on foreign AI infrastructure).

Drawing on recent surveys of economists and AI experts that warn of rising wealth inequality and a declining labour share of income under rapid automation (May; Tan), this essay argues that Singapore’s existing AI policy architecture—including SkillsFuture’s AI upskilling push and SME digitalisation and innovation grants—is necessary but still insufficient. Building on Singapore’s Budget 2026 initiatives and emerging AI governance standards, it proposes four main shifts.

First, the move from short-term solutions to an “AI-Ready Workforce” system, including a new AI Practitioner License for high-stakes sectors and a structured mid-career transition pathway for current workers. Second, from AI adoption only at the frontier to being commonplace for all firms through an “AI Commons for SMEs” platform and an AI Productivity Bond. Third, from voluntary guidelines to risk-tiered regulation, converting Singapore’s pre-existing AI Verify framework into a mandatory certification scheme and piloting an Algorithmic Audit Sandbox. Fourth, from mere dependence on foreign infrastructure to becoming a regional leader, position Singapore’s governance standards as the ASEAN benchmark through the Digital Economy Framework Agreement.

Singapore’s strength has always rested on managing complex trade-offs with structural changes. The AI era demands the same.

Main Essay

1. Background and Context

Singapore sits at an inflection point. Its trade-to-GDP ratio exceeds 300%, its financial sector contributes 13.8% of the economy (in 2023, according to Statista), and its ability to sustain productivity-led growth in an ageing society with very few natural resources has always depended on anticipating the next economic wave. Today, that wave is AI.

For this essay, “AI” refers to systems already in commercial deployment, such as machine learning models, large language models, computer vision, and robotics. Singapore has signalled AI’s centrality through the National AI Strategy 2.0 (NAIS 2.0) launched in December 2023, which organises adoption into 10 enablers, treating AI as economic infrastructure rather than speculative technology.

Expert economic evidence urges ambition moderated by caution. A 2025–26 survey of academic economists and other AI-focused experts around the world found that under a rapid AI progress scenario, economists forecast the U.S. GDP growth rising to around 3.5% per year, the labour force participation rate falling to about 55% by 2050, and the share of national wealth held by the top 10% of households rising to roughly 80% by 2050 — shifts that are historically large (Karger et al., 2026). For a society built on meritocracy and social cohesion, these are not abstract.

The central challenge, then, is this: AI can meaningfully raise Singapore’s competitiveness and productivity, but risks widening inequality, displacing workers in routine roles, and creating strategic vulnerabilities from technology dependence. Singapore’s AI-driven economic future should therefore focus on enhancing our workers, democratising AI tools across all firms, building trustworthy governance, and obtaining capable infrastructure.

2. Current Activities: How AI Helps and Where It Falls Short

2.1 Industry: Finance, Logistics, and Healthcare

Singapore's financial institutions are already among Asia's most advanced AI adopters. Banks use machine learning for fraud detection and anti-money laundering surveillance; MAS has issued sector-specific AI guidance and developed the Model AI Governance Framework alongside AI Verify (the world's first AI governance testing toolkit). However, hazy AI credit-scoring models risk disadvantaging SMEs without formal credit histories, exacerbating structural bias primarily at capital allocation, the automation of back-office analyst roles risks deskilling entry-level workers, narrowing career pathways in a sector that Singapore depends on for high-wage employment.

At PSA Corporation and Changi Airport Group, AI optimises most things from berth scheduling to predictive maintenance (PSA). SGTraDex, Singapore's supply chain data exchange, enables real-time data sharing across port, freight, and customs—a foundation that is exploited well by larger firms. Yet smaller manufacturers and carriers face high adoption costs, widening the productivity gap between the boundary and the rest of the economy.

In healthcare, SingHealth and NUHS have deployed AI diagnostic tools in radiology and emergency triage, while MOH's Health Systems Digital Transformation Blueprint sets system-wide objects. These are clear gains for an ageing society with a relatively constrained clinical workforce. However, AI tools trained predominantly on foreign datasets may not translate to reliability to Singapore's multiethnic population—an accuracy risk that current regulatory frameworks do not require to be addressed by firms. There is also a persistent risk of digital exclusion between elderly patients (whose care needs are generally the highest).

2.2 Workforce: Everyday AI Tools and Skills

AI productivity tools have expanded white-collar capabilities across Singapore's economy, through coding assistants, document summarisers, and multilingual translation engines that lower barriers to complex tasks for those already operating near the frontier. Singapore's AI Apprenticeship Programme (AIAP) has placed hundreds of AI engineers into companies on nine-month projects, and SkillsFuture provides course subsidies and S\$500 credits to all Singaporeans aged 25 and above.

However, these initiatives are more broad-based than targeted. AIAP, for example, reaches graduates entering the AI field; SkillsFuture's AI courses are often generic and unlinked to wage outcomes. Additionally, workers in predominantly

routine roles (such as clerical staff and basic data-entry operators) face high substitution risk, with the potential of causing mass-displacement. Evidence from comparable economies shows that early career workers in high-AI-exposure occupations have experienced measurable employment declines, with wages rising alongside employment falls as firms use fewer but more senior and skilled workers. Singapore's resident workforce, which is less mobile than foreign professionals, is most exposed to these increased structural pressures.

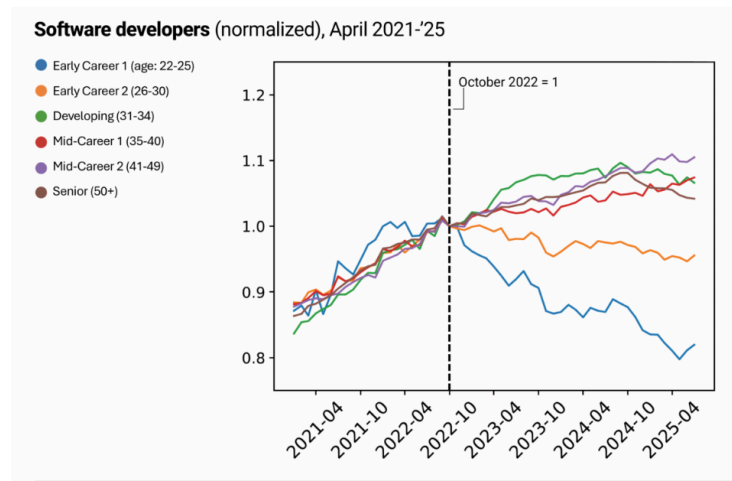


Fig 1. A graph showing the effect of AI on Software Developers' jobs after the mainstream launch of AI (especially with ChatGPT 3.5 in 2022). Early career workers have been hit hardest, while growth rates of other age groups have significantly reduced. (Antonello)

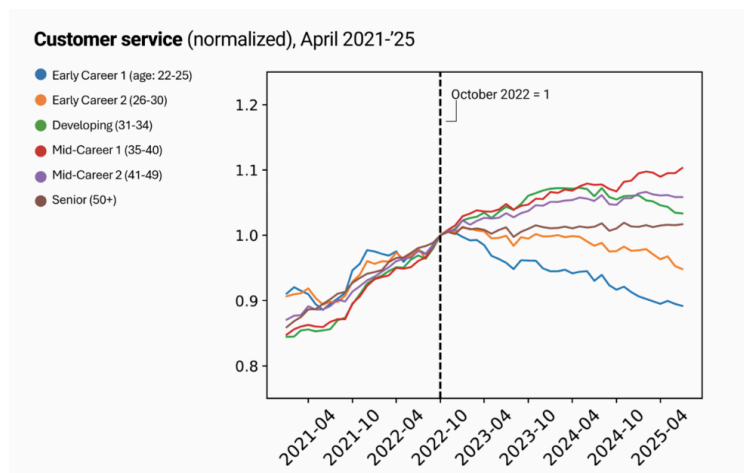


Fig 2. A similar graph displaying the effect of AI on the employment of Customer Service workers, which follow a more routine, repetitive role. (Antonello; Tae Won Kim)

2.3 Competitive Advantage: Singapore as an AI Hub

TSMC’s US\$11.6 billion advanced packaging investment, Nvidia’s regional headquarters expansion, and the presence of AWS, Microsoft Azure, and Google Cloud in Singapore as primary ASEAN cloud regions reflect clear geographical advantages. Nevertheless, Singapore’s AI capabilities depend heavily on imported frontier models, foreign cloud infrastructure, and external computers (such as how Singapore’s own SEA-LION AI is built on Gemma or Qwen, American and Chinese models respectively). This dependence is exacerbated by US-China technology decoupling and chip export controls. Singapore develops further limited frontier model capability domestically; its competitive advantage rests on application, deployment, and governance rather than model development. The benefits of Singapore’s AI boom might mostly go to big foreign firms and skilled foreign workers instead of local businesses and workers.

Fig 3.

~70%	SMEs’ share of Singapore’s workforce Yet SME AI adoption consistently lags large firms, capping economy-wide productivity gains (Shi Ning)
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Fig 4.

71.8% vs. 37.4%	Economist support for targeted AI retraining vs. UBI Expert consensus strongly favours targeted transition programmes over broad redistribution (Karger et al.)
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3. Evaluation of Current Activities

Three structural gaps undermine Singapore's current approach to AI adoption.

First, AI adoption is sectioned. SMEs employ roughly 70% of Singapore's resident workforce, but their uptake of AI tools consistently lags behind large firms and MNCs. EnterpriseSGs SME Digital Transformation Programmes have accelerated cloud adoption, but AI-specific use cases remain underdeveloped—something that reduces economy-wide productivity and undermines inclusive growth.

Secondly, the labour market is polarised. Workers with AI competencies gain productivity and wage advantages, while those without face displacement. The economists surveyed by Karger et al. (2026) strongly favour retraining of such professionals (71.8% support), over even universal basic income or job guarantees (37.4% and 13.7% respectively). Yet Singapore's existing upskilling lacks a dedicated mid-career AI transition pathway.

Third, governance is running slightly behind technology. The Model AI Governance Framework and AI Verify represent first-of-their-kind initiatives, but both are voluntary. High-risk AI deployments in credit scoring, medical diagnostics, and hiring currently face no mandatory requirements. As the gap between AI capability and regulation steadily widens, these risks accumulate; to address them, I propose the following four shifts.

4. Main Policy Shifts

4.1 Shift 1: From Unorganized Upskilling to an AI-Ready Workforce

Singapore should treat AI Capability as a key economic framework—as fundamental as broadband or primary education.

Three focused extensions of existing initiatives can be used to achieve this.

First, sector-specific AI credentials, which could be stackable qualifications co-designed by MAS, MOH, and IMDA with industry and IHLs for finance, healthcare, logistics, and public service—they should also be tied to wage increments and be considered horizontally portable across employers/firms. The Certis-SUTD AI Literacy Programme’s tiered design could provide a replicable template, with standardisation across providers added on.

More distinctively, Singapore should introduce a tiered AI Practitioner License for high-stakes sectors, similar to the Chartered Financial Analyst credential or Singapore Medical Council registration. Professionals deploying AI in regulated contexts would require a Foundation, Professional, or Expert license issued through a joint MAS-MOH-IMDA body and renewed through continuous participation. This creates accountability sector-wise, establishes trustworthy professional standards, and produces an internationally recognized credential.

Third, a structured mid or late-career AI Transition Pathway, funded through an extension of SkillsFuture Career Transition programmes, should serve workers displaced from at-risk clerical and operations roles. Participants would receive up to twelve months of training leave, a stipend, and guaranteed job-matching support into complementary functions. Evidence from adjustment programmes in comparable economies confirms that well-designed transition support lowers unemployment and generates higher returns.

Forecasts of Labor Force Participation Rate

Lines show medians of 50th percentile forecasts across participants. Shaded regions span the median 10th to median 90th percentile forecasts.

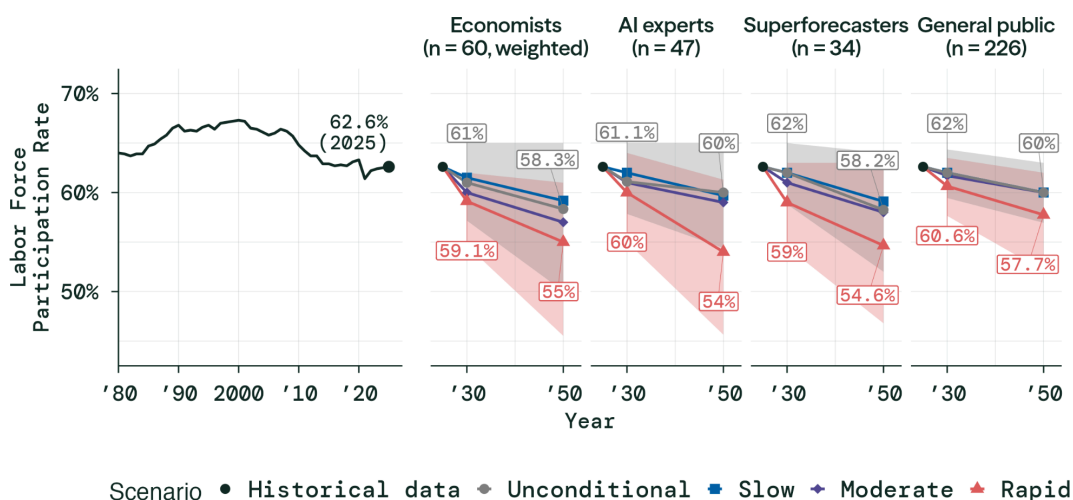


Fig 5. AI progress scenarios vs economic forecasts for 2030 and 2050, showing economists expect a large loss of workers in the labour force over the next 20 years. (Forecasting Research Institute)

4.2 Shift 2: From AI at the Frontier to AI for All

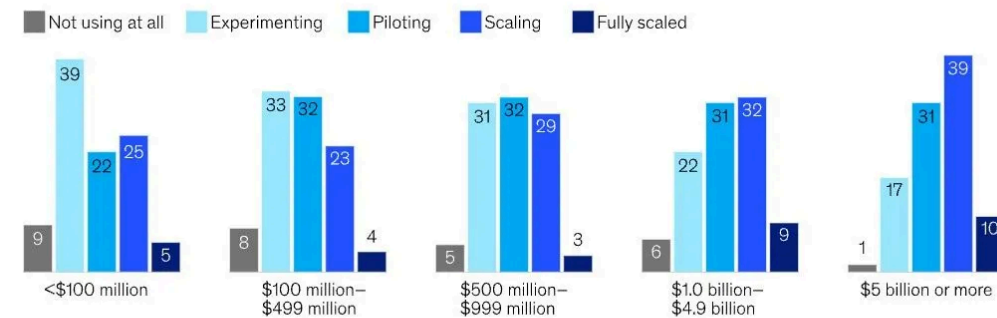
Singapore cannot afford an AI transition that forgoes its SMEs. Two structural innovations may extend AI's research across the economy.

First, an AI Commons for SMEs—operated by IMDA or GovTech—should offer pre-approved, sector-relevant AI models accessible through Corppass with standardised onboarding and subsidised pricing. By integrating with the Peppol e-invoicing network and SGTraDex, the platform lowers switching costs and maximises interoperability for SMEs.

Second, EnterpriseSG's existing AI adoption grant framework should incorporate an AI Productivity Bond: SMEs that adopt approved AI tools and show clear productivity or wage improvements — measured using IRAS corporate tax returns or CPF wage records that the government already holds — can receive subsidised loans at lower interest rates from EDB-linked financiers. This outcome-based approach replaces one-off software purchases with proven impact, and uses Singapore's existing government data systems as a cheap, built-in way to track and verify results.

Larger companies lead the way in scaling AI beyond pilots.

Phase of organization's use of AI, by company revenues,¹ % of respondents



Note: Figures may not sum to 100%, because of rounding.

¹Respondents who said "don't know" are not shown, but represent <2% of the total.

Source: McKinsey Global Survey on the state of AI, 1,993 participants at all levels of the organization, June 25–July 29, 2025

McKinsey & Company

Fig 6. AI adoption sharply increases with company size, leaving most small firms stuck in pilot or non-use stages.

(McKinsey & Company)

4.3 Shift 3: From Voluntary Guidelines to Risk-Tiered Governance

Singapore's Model AI Governance Framework and AI Verify are globally respected but voluntary. The EU AI Act has moved to mandatory obligations tied to risk categories, with US sectoral rules trending the same way. Singapore must move ahead of this curve rather than react to it.

A risk-tiered framework should preserve the non-intrusive rules for low-risk applications, while imposing mandatory explainability standards, human oversight requirements, and periodic independent audits for high-risk usage. Crucially Singapore should establish an Algorithmic Audit Sandbox, extending the MAS FinTech Regulatory Sandbox model where firms deploying high-risk AI would test systems under regulator-approved evaluation metrics before live deployment; this would make it a trusted jurisdiction for responsible AI in ASEAN.

At the same time, A.I. Verify should be turned into a clear ranking system for AI safety—a "Trusted AI" label that other countries would recognize. Singapore has led the way in banking rules before; it should do the same now for how AI is managed.

4.4 Shift 4: From AI Dependence to Resilient Infrastructure and Regional Leadership

Singapore's compute dependency is a vulnerability. US chip export controls, cloud concentration among three hyperscalers, and limited domestic frontier model capability represent risks that NAIS 2.0's Compute pillar must actively address: through diversified cloud provider incentive, energy efficient green data centers, and regional compute partnerships.

Crucially, Singapore should not aspire to self-sufficiency across the AI stack—that is neither feasible nor efficient for a small, open economy. Instead, it should target and build off its existing strengths: multilingual AI for Southeast Asian languages (using SEA LION as a platform), maritime and port logistics optimisation, city infrastructure management, and financial compliance.

Internationally, Singapore's most powerful lever is the sharing of governance across nations. Through the ASEAN DEFA and bilateral digital economy agreements, Singapore should promote the adoption of AI Verify standards and risk-tiered governance across ASEAN—similar to precedents like the PayNow-UPI cross-border corridor. Singapore's comparative advantage lies in trusted institutions, reliable rules, and the willingness to move first.

5. Conclusion

Artificial intelligence will reshape Singapore's economy regardless of the choices made today. What remains open is the distribution of its gains and costs. Expert economists are clear: even under moderate AI progress, inequality rises and labour's share of output falls without corrective policy. Singapore has always been a small state, giving it the ability to manage technological transitions more deliberately than larger ones could afford to.

To keep growth inclusive, Singapore must equip workers with recognised AI skills, ensure SMEs share in productivity gains, and regulate AI with a clear, risk-based governance system. If it does this while leading the region from a position of principle rather than dependence, AI becomes the next stage of Singapore's economic model instead of a threat to it.

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